A bond is a schedule of promised cash flows

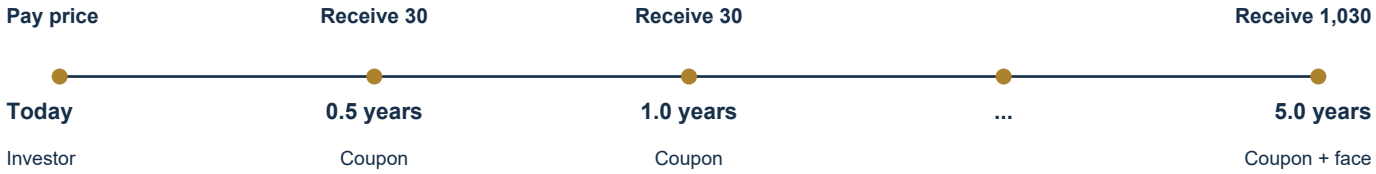
LECTURE 7 · REVIEW NOTES

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When we price a bond, I want you to start with the contract rather than a formula. Identify face value, the coupon and the payment dates. Then ask what those promised cash flows are worth today. Face value is the principal repaid at maturity; price is what an investor pays for the entire payment stream. Those amounts need not be equal.

Unless stated otherwise, yields and coupon rates here are quoted annually with semiannual compounding and payments. A USD 1,000 bond with a 6% coupon pays USD 30 every six months. Five years creates ten payment periods.

$$C = \frac{FV \times c}{2}, \qquad n = 2T$$



From the issuer's perspective, the signs reverse: receive issue proceeds today, pay coupons and principal later. Issuing USD 1,000 of face value does not guarantee USD 1,000 of proceeds. Market price determines the amount raised.

Coupon rate and yield answer different questions

The coupon rate specifies contractual interest relative to face value. Yield to maturity is the single discount rate that equates the present value of promised payments to the observed price. Once a fixed-rate bond is issued, the coupon stays fixed while price and yield can change.

Pricing on a coupon date

These examples value bonds on coupon dates, with zero accrued interest. Between coupon dates, distinguish the quoted clean price from the full price paid, which includes accrued interest.

Discount each cash flow, or use the annuity formula

For a level-coupon bond, discount every coupon and the face value using the periodic yield. If the annual quoted yield is y and compounding is semiannual, the rate per period is y / 2.

$$P = \sum_{k=1}^{n} \frac{C}{(1 + y/2)^k} + \frac{FV}{(1 + y/2)^n}$$

Coupons form an annuity. Grouping them gives the same price with fewer calculations. The first term values the coupon stream and the second values principal. Do not include the final coupon twice.

$$P = C\frac{1 - (1 + y/2)^{-n}}{y/2} + FV(1 + y/2)^{-n}$$

Five-year bond: FV = 1,000; coupon = 6%	Price
Required annual yield = 4%	1,089.83
Required annual yield = 6%	1,000.00
Required annual yield = 8%	918.89

At a 6% yield, the coupon is exactly what the market requires, so price equals face. At a 4% yield, the 6% coupon is attractive and price exceeds face: a premium. At an 8% yield, the coupon is insufficient at par and price falls below face: a discount.

Price and yield move in opposite directions

Higher required returns reduce the present value of fixed promised payments. Longer maturities generally increase sensitivity because more value comes from distant cash flows. Credit deterioration can also lower price by increasing the return investors demand.

One YTM versus a curve of zero rates

YTM discounts all payments at one rate. A zero-rate curve discounts each payment using its maturity-specific rate. Both can describe the same price, but YTM summarizes that price rather than replacing the entire curve.

Solve for YTM and interpret it

When price is known, solve the pricing equation for yield. Excel RATE returns a rate per payment period. With semiannual coupons, multiply by two to report annual nominal YTM. Use opposite signs for the price paid and cash received.

$$y = 2 \times \mathrm{RATE}(n, C, -P, FV)$$

Face 1,000; annual coupon 15%	2-year bond	10-year bond
Price	1,200	1,500
Semiannual coupon	75	75
Number of periods	4	20
Annual nominal YTM	4.4390%	7.7296%

Both bonds trade at premiums and yield less than their 15% coupon. Investors receive high coupons but pay more than the principal ultimately returned. YTM includes that difference and payment timing. Current yield, coupon divided by price, does not.

Compare yields on a consistent basis

An illustrative 30-year corporate bond with face 100, coupon 4.45% and price 93.779 has a YTM of 4.8455%. Against a comparable 4.00% Treasury yield, the spread is approximately 84.5 basis points. One basis point is 0.01 percentage point.

Hold currency, maturity and convention reasonably constant. Spreads can reflect credit risk, liquidity and embedded options. Credit ratings summarize credit assessments; they are not repayment guarantees. Even a low-credit-risk Treasury retains interest-rate risk.

Promised yield is not a guaranteed return

Default can prevent promised payments. Selling before maturity exposes the investor to a changing sale price. Coupon reinvestment rates affect terminal wealth. A distressed bond's high quoted YTM is therefore not evidence of an easy high return.

Policy rates are only part of the story

Central banks target short-term rates. Longer-term yields also reflect expectations and risk premiums. Mortgage and corporate rates add credit, liquidity and contract features; they need not move one-for-one with the policy rate.